

Question number	Answer	Mark
9(a)(i)	The only correct answer is B <i>A is incorrect because Q is the strict osmoconformer</i> <i>C is incorrect because R is the osmoregulator</i> <i>D is incorrect because Q is the strict osmoconformer</i>	(1)

Question number	Answer	Additional guidance	Mark
9(a)(ii)	An explanation that makes reference to three of the following points: - so that {no (net) / little / not much} <u>osmosis</u> (of water) will occur (1) (do not want water to leave) otherwise cell will {dehydrate / shrink} (1) (do not want water to enter) otherwise cell will {swell / burst} (1) {energy / ATP} is not used in regulating solute concentrations (1)	ACCEPT no movement (of water) as no <u>osmotic effect</u> IGNORE refs to water concentration IGNORE flaccid IGNORE turgid / plasmolysis NB. eg 'no risk of cells bursting due to <u>osmosis</u>' = 2 marks	(3)

Question number	Answer	Additional guidance	Mark
9(b)(i)	any (whole number) value from 550 to 580 (1)	NB check table if no answer given if answer in table is different to answer on the answer line then mark the answer on the answer line	(1)

Question number	Answer	Additional guidance	Mark
9(b)(ii)	- correct calculation (1) 500.00 / 142.86 / 83.33 (1)	$((54 - 9) \div 9) \times 100 = 500$ $((54 - 9) \div 54) \times 100 = 83.333333$ $(45 \div 31.5) \times 100 = 142.8571$ DO NOT ACCEPT recurring numbers Correct bald answer scores 2 marks Bald answer of any correct value but incorrect rounding scores 1 mark	(2)
Question number	Answer	Additional guidance	Mark

9(b)(iii)	An explanation that makes reference to the following points: - because the total concentration of all the ions affects the water potential (1) (total) ion concentration (for each organism) is similar to that of the sea water (1)	ACCEPT solute concentration / solute potential / osmotic potential IGNORE water concentration	(2)
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Question number	Answer	Additional guidance	Mark
9(b)(iv)	because: - their food contains different concentrations of each ion - they eat different food - they incorporate different concentrations of each ion (into other molecules) - it depends on the requirements of the {cells / animal / body} - they absorb ions differently - they live in different areas of the sea - different parts of the sea have different ion concentrations	IGNORE refs to size of animal	(1)

Question number	Answer	Additional guidance	Mark
9(b)(v)	An explanation that makes reference to the following points: • active transport (of ions) (1) • because some of the ions have a higher concentration inside the {organisms / cells / tissues} (than in the sea water) (1)	DO NOT ACCEPT {diffusion / facilitated diffusion / osmosis} ACCEPT the converse to move the ions {against / up} the concentration gradient to move ions from a low to high concentration (of ions)	(2)